Can you trust an AI equity research report? Auditing and improving FinRobot's equity pipeline

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> I ran FinRobot's `finrobot_equity` pipeline on ten companies and audited three things: the numbers, the valuation, and the prose. The reported financials are transcribed correctly. The valuation engine, however, reduces to one constant multiple (10.84x FY+3 EBITDA per share for every company), and ten of its sixteen assumptions are not shown. In the prose, 18% of claims are contradicted by the data the model was given, and two thirds of those contradictions are period or status errors: reported results called "projected" or "guidance", or the historical CAGR attached to forecast years. The cause is the evidence format, not the model inventing numbers. Labelling the evidence and adding a verifier cut contradicted claims from 16.3% to 1.3% across ten companies, with the same result on five held-out names. Most of the gain comes from labelling the evidence.

1. Setup

System. `finrobot_equity` fetches statements, ratios and peers from Financial Modeling Prep (FMP), builds a three-year forecast, generates eight text sections with an LLM, and renders an HTML report. Three facts about the code path that actually runs shaped the audit:

- **The eight-agent team is not used.** The CLI calls `text_generator_agents.generate_text_section`, a single Chat Completions call per section (temperature 0.7, max 1,000 tokens). `EquityResearchAgentManager` is never instantiated. The evidence for every section is therefore `_prepare_user_prompt`: three tables (financial metrics and forecast, peer EBITDA, peer EV/EBITDA), with news only for the news section.
- **It does not run with a new FMP key.** FMP serves its legacy `/api/v3` and `/api/v4` endpoints only to pre-August-2025 subscribers, so all 13 endpoints return 403. I moved them to `/stable/`, mapped renamed fields back, and stopped error messages from printing the API key (patch 0001).
- **Failures are silent.** Any API error yields canned text, and `major_takeaways` is replaced by canned text whenever the model omits the exact string "Revenue Growth:", which happened in 14 of 14 runs. Step 2 regenerates two sections with `gpt-4o-mini` regardless of the configured model.

Component	Choice
Code	FinRobot `2717499b` + patch 0001 (FMP `/stable/`) + patch 0002 (Part 3)
Data	FMP free plan: 250 calls/day, no news endpoint, symbol whitelist; SEC EDGAR XBRL for the data audit; Yahoo Finance for FY-end prices and analyst consensus
Models	Writer `gpt-4.1` (T=0.7); Step 2 regeneration `gpt-4o-mini`; audit judge `gpt-4.1` (T=0); verifier `gpt-4.1-mini` (T=0)
Cost	About $9.6 in total: grounding judge $7.03 (recorded tokens), Part 3 writer ~$1.29, verifier $0.39, pipeline runs ~$0.90; FMP free
Runtime	42–63 s per report (Step 1: 26–43 s, Step 2: 15–29 s); scoring 438 claims: about 12 minutes

2. Coverage universe

Set	Company (peers)	3-year price	Why it is in the set
Starter	NVDA (AMD, INTC)	+402%	Required starting set. All five rose over three years; INTC mostly in the last year (+177%).
	AMD (NVDA, INTC)	+458%	
	INTC (NVDA, AMD)	+192%	
	AAPL (MSFT, GOOGL)	+91%	
	GOOGL (MSFT, META)	+157%	
Healthcare	ABBV (PFE, JNJ)	+106%	Winner
	JNJ (PFE, ABBV)	+80%	Winner
	PFE (JNJ, ABBV)	+3%	Laggard
	UNH (JNJ, PFE)	−23%	Laggard; managed care, no SG&A line in FMP data
	MRNA (PFE, JNJ)	+115%	Turnaround (+717% in one year after a collapse); loss-making, R&D at 161% of revenue

Healthcare was chosen because R&D-heavy companies stress the forecast (which omits R&D) and include a loss-maker. The FMP free plan does not cover LLY, MRK, BMY, AMGN, GILD or any managed-care company, so UNH is compared with JNJ and PFE. The five healthcare names were not used while designing or tuning the Part 3 changes (the pilots used NVDA and AMD) and serve as a held-out set.

3. Audit results

3.1 Data audit: do the numbers match the filings?

For NVDA and AAPL, 27 line items were checked against the 10-K XBRL facts. Each filing value carries its accession number (`audit/data_audit.csv`). 18 match within rounding, and every difference has an explanation:

Item	Pipeline	Filing	Verdict and explanation
NVDA EBITDA FY2026	$144.6bn	$133.2bn	Definition: FMP uses pre-tax income + interest + D&A, so it counts $11.1bn of non-operating income (mostly investment gains). The report never says so.
NVDA EV/EBITDA FY2026	31.4x	34.1x	Definition: the same EBITDA, plus lease liabilities counted as debt.
NVDA EPS FY2024	$1.21	$12.05	Restated for the 10-for-1 split of June 2024. Correct, but undocumented.
AAPL EBITDA FY2025	$144.4bn	$144.7bn	Same definition difference; immaterial for Apple (−0.2%).
NVDA revenue "2025E"	$226.7bn	$130.5bn	Mislabelled period: forecast columns are hard-coded 2025E–2027E, so "2025E" is FY2027.
AAPL revenue "2025E"	$437.0bn	$416.2bn	Mislabelled period: the table shows FY2025 twice.

The data layer is not the problem; labels and definitions are. Net income and R&D (8–9% of revenue) never reach the text sections at all.

3.2 Forecast and valuation: are the assumptions defensible?

Forecast. The defaults (5% / 6% / 4% growth, +1pp gross margin, −0.5pp SG&A margin a year) apply to every company. Next-year revenue comes out 9% (AAPL) to 45% (NVDA) below analyst consensus. Forecast EBITDA is

computed as gross profit − SG&A, so R&D disappears in the first forecast year. AMD's EBITDA margin jumps from 21.0% to 39.0%, MRNA's loss of $2.6bn becomes a profit, and UNH, which has no SG&A line, gets no forecast EBITDA at all. As revised assumptions I used consensus growth for FY+1 and FY+2, faded halfway to 5% for FY+3, with flat margins (`forecast_assumptions.csv` lists the CLI flags).

Valuation. I reproduced NVIDIA's $82.24 target by hand to the cent. The register of its assumptions (`valuation_assumptions.csv`) shows ten of sixteen are not visible in the report: net debt is set to 10% of EV, FCF to 60% of EBITDA, WACC to 10%, and the EV/EBITDA multiple is a 12x default printed as "Historical Avg Multiple". The peer method is dropped silently because Step 2 passes peer data in the wrong shape. With the expected input it would have returned $212.79. Both remaining methods are linear in one input, so for every company **target = 10.84 × FY+3 EBITDA / shares**.

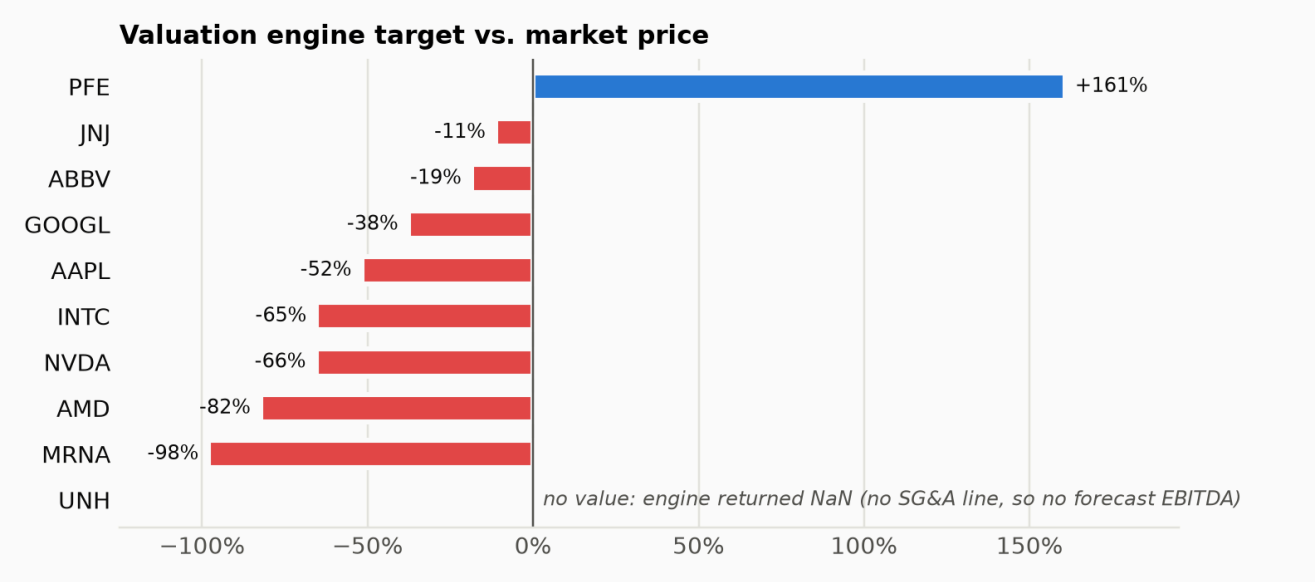


Figure 1. Engine target vs. market price. The sign and size of the "implied move" track the engine's constants, not the companies. PFE's +161% comes from forecast EBITDA that omits R&D. UNH has no bar: FMP reports no SG&A line for it, so the forecast EBITDA (gross profit − SG&A) is empty, the engine returns NaN, and the report's valuation section is silently blank ("completed with 0 methods").

Sensitivity. ±5% revenue moves the target exactly ±5%, and ±2pp of margin moves it 3–7%. By the assignment's test the point estimate is therefore not "theater", but only because it is linear. Choices the reader never sees move it far more: consensus instead of default growth moves it +6% (AAPL) to +251% (NVDA); each company's own historical EV/EBITDA instead of 12x moves it +28% to +179%. The shipped sensitivity module never touches the target price, and its "95% confidence interval" is a hard-coded ±29.4% for every company.

3.3 Grounding: is the prose supported by the evidence?

Method. `audit/grounding.py` rebuilds the exact prompt each section was written from. A `gpt-4.1` judge splits the section into atomic claims and labels each Supported, Unsupported or Contradicted, citing the cells it used and the year and status (reported vs. forecast) the claim asserts. Deterministic rules then override the judge where the cited cells and the claim's wording conflict: a reported value worded as a forecast, or the CAGR attributed to forecast years. A number check looks up every figure in the tables. Of a stratified sample of 65 claims, the scorer agrees with the hand review on 59 (91%). All 30 rule overrides were correct. The judge is lenient: 4 of 15 randomly sampled "Supported" claims were not.

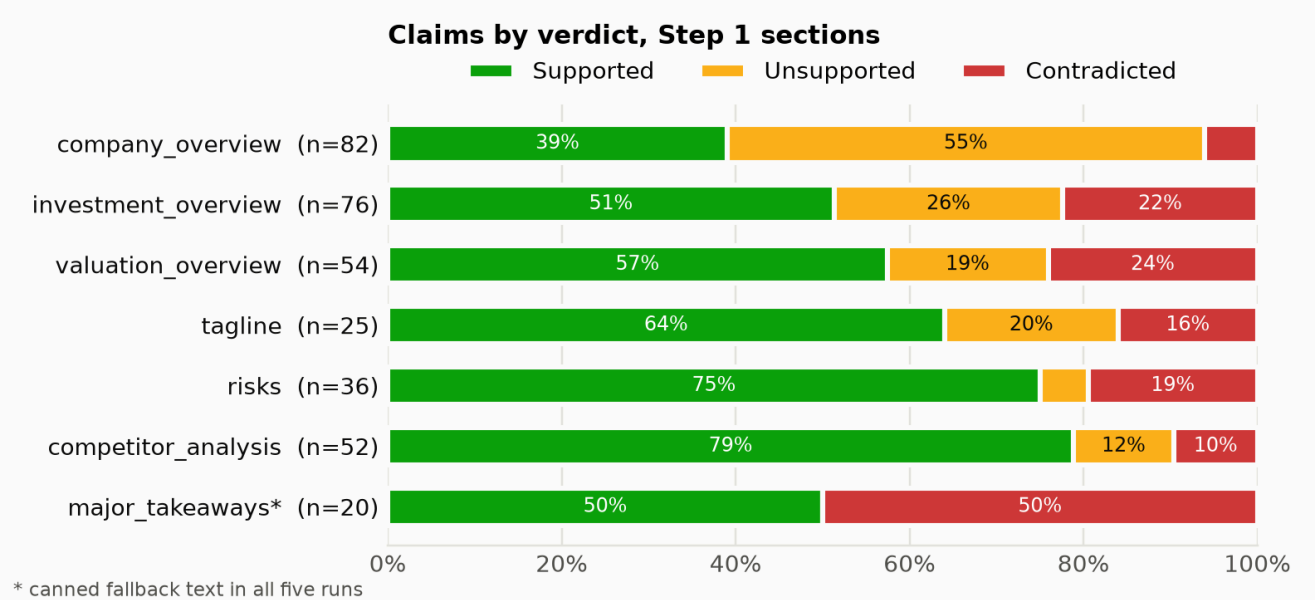


Figure 2. Verdicts for 345 claims in the Step 1 sections of the five starter companies. Corrected for the judge's measured leniency, the overall grounding rate is about 44% rather than the measured 57%.

Which sections hallucinate most, and why it is structural.

- **Unsupported claims follow the gap between what a section is asked for and what its prompt contains.** `company_overview` is asked for "business model, products/services, market position" and given three financial tables. 55% of its claims are unsupported: founding year, headquarters, product lines, market position. Overall, quantitative claims are 75% supported and 1% unsupported; qualitative claims are 28% supported and 63% unsupported.
- **Contradictions come from the evidence format.** 23% of quantitative claims are contradicted. Of the 79 contradictions, 31 present a reported figure as projected, expected or guidance, and 22 attach the historical CAGR to forecast years. The prompt has no date, so the model writes from its training cutoff: one section is headed "Investment Update – June 2024", and upstream patches only the title. Forecast columns labelled 2025E sit next to 2025A/2026A, the CAGR column does not say which years it covers, and AMD's 2027 EBITDA appears twice with different values. Sections that ask for an "outlook" or "fair value" have the highest contradiction rates (22–24%).
- `risks` **does not behave as the assignment predicts**, because the 600–800-word risks agent is never called. The CLI asks for "5 key investment risks, concise". The model anchors each bullet on a table figure and phrases the rest as hypotheticals, which are not checkable claims. That gives a high grounding rate (75%) but not risk analysis.
- **Step 2 makes two sections worse.** The gpt-4o-mini rewrite of `competitor_analysis` is 45% grounded against 79% for the gpt-4.1 version it replaces.

3.4 Stability: same input, three runs

Output	Run 1	Run 2	Run 3	Stable?
Rating / target / engine target	Buy / $339.17 / $82.24	same	same	Yes, because no LLM produces them
Top-3 risk categories	valuation, growth, margins	valuation, growth, competition	valuation, growth, competition	No (Jaccard 0.5 / 0.5 / 1.0; all five categories identical)
Grounding rate (Step 1)	43%	44%	46%	In aggregate, yes
Tagline grounding	100%	75%	50%	No; section-level results swing up to 100 points

The error *rate* is a property of the prompt and evidence; which sentence is wrong is chance. That argues for fixing the prompt, not individual outputs, and for evaluating on many samples.

4. Improvement: labelled evidence and a verifier (Option A)

Design (patch 0002: 4 files, +289/−20 lines, 5 unit tests).

1. **Evidence.** Forecast years are labelled from the base year (FY+1..FY+3). Every writer prompt starts with reading notes: the report date, which columns are reported and which are the pipeline's forecast (with its assumptions), the CAGR period, the projected peer year, and that the data has no guidance, consensus or news.
2. **Verifier** (`--enable-verifier`, `gpt-4.1-mini`). After each section is written, it checks the section against the writer's data plus the notes and replaces sentences that mislabel reported vs. forecast figures, misattribute the CAGR, misquote numbers or cite non-existent sources. Changes are exact sentence replacements, logged in `verifier_log.json`. Statements the data does not cover are listed, not deleted. A verifier error never blocks the report.
3. The takeaways check no longer discards valid output (it matched "Revenue Growth:" exactly).

Evaluation. A 2 × 2 experiment on the same saved data: 10 companies × 7 sections, original vs. labelled prompt, each with and without the verifier. The writer is called exactly as the pipeline calls it. Each text is scored against the prompt its writer saw, using the Part 2.3 scorer plus an LLM-free status-error counter. The counter shows 0 false positives and detects 50/50 injected errors; its precision on real text is about 85%.

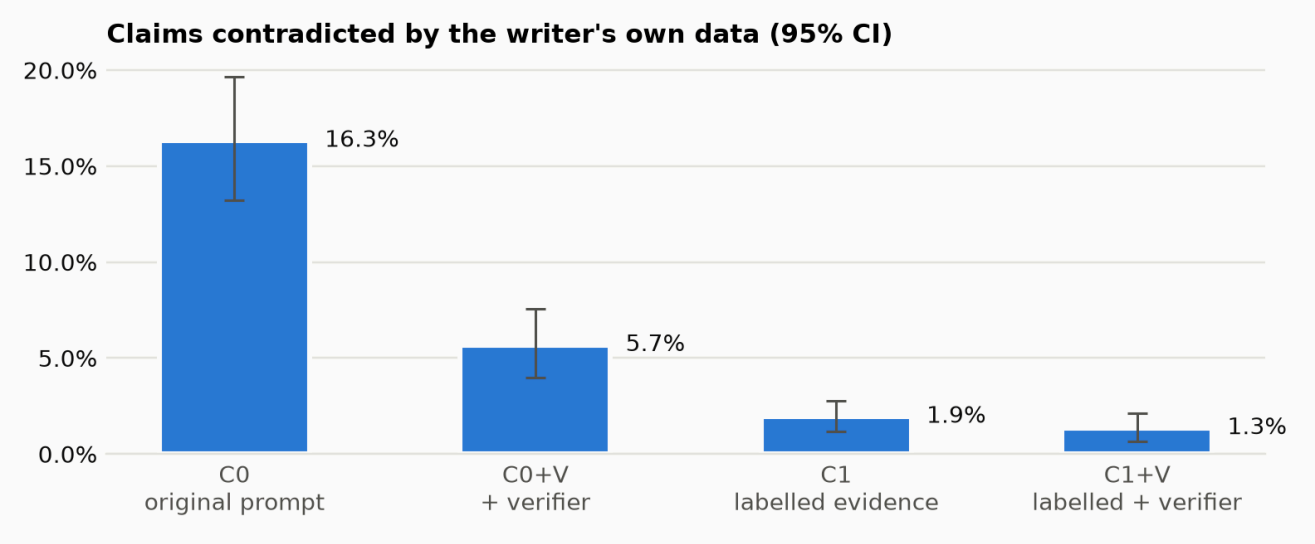


Figure 3. Share of claims contradicted by the writer's own data, 10 companies × 7 sections per condition; 95% bootstrap intervals over company-sections.

Metric	C0 original	C0 + verifier	C1 labelled	C1 + verifier
Contradicted claims	16.3%	5.7%	1.9%	1.3%
— held-out healthcare only	16.6%	6.6%	1.5%	1.3%
Grounding rate	57.6%	68.4%	82.5%	84.1%
Unsupported claims	26.1%	25.9%	15.5%	14.6%
Supported claims (count)	413	506	680	686
Status errors per section (LLM-free counter)	1.47	0.27	0.03	0.03
Words per section	227	224	244	240

- **Labelling the evidence does most of the work**, and it does not trade content for accuracy: supported claims rise by two thirds and sections get slightly longer.

- **The verifier helps only with the context the writer lacked.** Without the report date it repeated the writer's error: it rewrote "consensus estimates project revenues to climb to $130.5 billion in 2025" as "revenues are projected to climb to $130.5 billion in 2025". On top of labelled evidence it adds little (1.9% → 1.3%, intervals overlap), so it is a cheap safety net (one mini call per section) rather than the fix.
- **What it still gets wrong.**
  - Unsupported background claims remain (15%), because the overview still asks for content the data lacks.
  - 44% of the verifier's proposed corrections are dropped because the model did not quote the sentence verbatim, and some corrections are noise.
  - Step 2's gpt-4o-mini rewrite bypasses both fixes.
  - Forecast EBITDA still omits R&D, and the valuation engine is unchanged.

5. Judgment: would I hand this to a portfolio manager?

Part of the report	Today	Why, and what would have to change
Reported financial tables	Usable with caveats	Transcribed correctly; the EBITDA definition must be stated, and net income and R&D should be shown.
Forecast table	No	One set of constants for every company; R&D omitted from EBITDA. Needs company-specific drivers (consensus or a model) and the full cost structure.
Rating and target price	No	The page-1 rating is the analyst consensus restated, while the report's own valuation says −11% to −98% (or +161% for PFE). It needs one valuation with disclosed WACC, multiple and net debt, and the rating should follow from it.
Numerical prose	With patch 0002, close	Contradictions fall from 16% to about 1%. It still needs a human read of anything tied to a decision.
Descriptive prose (overview, competition)	No	12–55% of claims come from model memory. It needs source documents (filings text, segment data) in the evidence, or a narrower brief.
Risks and news	No	Risks restate valuation figures; no news on the data plan. They need risk-relevant evidence (filing risk factors, news).

The pipeline is a good drafting tool for the numerical description of reported results once the evidence is labelled. It is not yet a research product: the parts a PM would act on (forecast, valuation, rating) are fixed constants presented as analysis, and the report contradicts itself between page 1 and its valuation section.

So what is it good for?

- **As a first draft, not a verdict.** In about a minute and for under $0.10, it describes reported results correctly, and with patch 0002 about 1% of its numerical claims contradict the data. That is useful for drafting earnings updates across many names, or for covering names no analyst follows. The condition is that the rating, target and forecast are removed, or replaced by assumptions the PM sets, so the output becomes a labelled data brief.
- **As a map of where to look.** The audit turns a uniformly confident document into one with calibrated trust. Reported tables and numerical description can be relied on; forecast, valuation, rating and descriptive claims need a human. Review time can go where the risk is.
- **As a test harness.** The grounding scorer and the LLM-free counter apply to any system that writes from structured data, and they make prompt or model changes measurable before they ship.

To support a PM's decision directly, four things would have to change:

1. company-specific forecast drivers, with the full cost structure;
2. a valuation with disclosed WACC, multiples and net debt, plus sensitivity;
3. source documents and citations for descriptive claims;
4. the verifier as a gate before publication, with a human signing off on any rating.

6. Limitations of this study

- The judge and the writer are the same model family. The judge was measured at 91% agreement on 65 hand-reviewed claims and errs towards leniency, so grounding rates are, if anything, optimistic. In the hand review, Claude reviewed all 65 claims first and the author independently adjudicated the six disagreements.
- One sample per company-section and condition in Part 3. Section-level results vary a lot between runs (§3.4), so only pooled results (70 company-sections per condition) are reported.
- Coverage is limited by the FMP free plan: no news, a symbol whitelist, and no managed-care peers for UNH.
- The LLM-free counter's precision rules (clause boundaries, negation, explicit "A" labels, the starting point of a change, the position of forecast wording) were refined while inspecting the Part 3 texts, including the held-out names. It is therefore used only to corroborate. The headline result rests on the judge and its rules, which were fixed during the Part 2.3 audit, before Part 3, and not tuned on those texts.
- The data audit covers two companies (27 line items). The valuation was reproduced for one company and generalised through the closed form, which was checked on all five starter companies.

All tables, notebooks, patches and commands are in this folder; see `README.md`. Every number in this report is produced by the notebooks in `audit/`.